

4	(a) (i) force proportional to product of (two) charges and inversely proportional to square of separation reference to point charges	M1 A1	[2]
	(ii) $F = 2 \times (1.6 \times 10^{-19})^2 / \{4\pi \times 8.85 \times 10^{-12} \times (20 \times 10^{-6})^2\}$ $= 1.15 \times 10^{-18} \text{ N}$	C1 A1	[2]
	(b) (i) force per unit charge on <i>either</i> a stationary charge or a positive charge	M1 A1	[2]
	(ii) 1. electric field is a vector quantity electric fields are in opposite directions charges repel <i>Any two of the above, 1 each</i>	B2	[2]
	2. graph: line always between given lines crosses x-axis between $11.0 \text{ } \mu\text{m}$ and $12.3 \text{ } \mu\text{m}$ reasonable shape for curve	M1 A1 A1	[3]